Case Study: Working with Two Lists Using Java Stream API

Problem Statement

You are given two lists: **List1** and **List2**. Use the Java Stream API to perform the following operations.

Tasks

1. Calculate Aggregate Values
Find the sum, average, minimum, and maximum of List1 and List2 separately.

2. Find the K Most Frequent Elements
Find the K most frequent elements in List1 and List2 separately. The value of K should be provided by the user.

3. Find the Second Most Frequent Element
Find the second most frequent element in List1 and List2 separately.

4. Merge the Two Lists
Merge List1 and List2 into a single list using flatMap().

5. Find Common Elements
Find the elements that are common to both List1 and List2.

6. Map Strings to Their Lengths
Create a list of streams and map each string to its length, handling the two lists separately.

7. Find Prime Numbers
Find and display all prime numbers in List1 and List2 separately.

8. Find Non-Prime Numbers
Find and display all non-prime numbers in List1 and List2 separately.

9. Find Repeated Strings
Given two lists of strings, find the strings that occur more than once in each list separately.

10. Find the First Repeating Element
Find the first repeating element in List1 and List2 separately.

11. Find the Longest Word
Find the longest word in List1 and List2 separately.

12. Find the Smallest Word
Find the shortest word in List1 and List2 separately.

Implementation Note

Use two integer lists for the numerical operations (Tasks 1–5 and 7–8), and two string lists for the string-based operations (Tasks 6 and 9–12). For Task 4, merge the two integer lists using flatMap().